

Confidence interval for difference of two means; independent samples, variances unknown but assumed to be equal

Apples example

NY apples	LA apples		NY	LA	
\$ 3.80	\$ 3.02	Sample mean	\$ 3.94	\$ 3.25	
\$ 3.76	\$ 3.22	Sample std	\$ 0.18	\$ 0.27	
\$ 3.87	\$ 3.24	Sample size	10	8	
\$ 3.99	\$ 3.02				
\$ 4.02	\$ 3.06	Pooled variance	0.05		
\$ 4.25	\$ 3.15	Pooled std	0.22		
\$ 4.13	\$ 3.81				
\$ 3.98	\$ 3.44	95% CI, $t_{16,0.025}$	2.12		
\$ 3.99					
\$ 3.62					

Confidence intervals

T	CI low	CI high
95%	\$ 0.47	\$ 0.92

Pooled Variance t-interval.

The confidence interval is calculated as:

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \cdot \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Where:

- $\bar{x}_1$ and $\bar{x}_2$ are the sample means.
- n_1 and n_2 are the sample sizes.
- t^* is the critical value from the t -distribution table.
- s_p^2 is the **pooled variance**.

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

two sample variances (s_1^2 and s_2^2):

the degrees of freedom used to find the t^* critical value is:

$$df = n_1 + n_2 - 2$$